

ARACAJU, Brazil

WMO: 830950

Lat: 10.984S

Lon: 37.070W

Elev: 23

StdP: 14.68

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	69.7	71.3	62.5	84.7	77.6	64.1	89.6	77.4	19.7	76.7	18.5	77.6	3.8	310	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	9.4	89.8	79.8	88.7	79.4	87.9	79.1	80.9	87.3	80.2	86.4	79.9	85.9	12.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	151.0	85.0	78.7	149.1	84.9	77.6	143.4	84.1	44.6	87.7	44.0	87.0	43.4	86.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	17.2	16.1	DB	66.3	91.8	1.3	4.3	65.3	94.8	64.6	97.3	63.8	99.7	62.9	102.8	(4)
(5)			WB	64.1	82.2	2.6	2.3	62.3	83.9	60.7	85.2	59.3	86.5	57.4	88.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.1	82.1	82.6	82.5	81.7	79.9	78.4	77.3	76.9	78.2	79.8	80.6	81.6	(6)	
	DBStd	2.66	1.72	1.98	1.86	2.02	1.87	1.78	1.80	1.86	1.80	1.85	1.55	1.54	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	10992	996	912	1008	951	927	852	845	834	847	924	917	979	(10)	
	CDD65	5517	531	492	543	501	462	402	380	369	397	459	467	514	(11)	
	CDH74	55165	6025	5855	6574	5784	4715	3424	2761	2572	3151	4218	4597	5489	(12)	
(13)	CDH80	14174	1902	2072	2383	1905	1181	508	257	218	401	805	1027	1515	(13)	
(14)	Wind	WSAvg	9.5	9.9	9.7	9.1	8.3	7.9	8.5	9.4	9.5	10.3	10.6	10.6	10.3	(14)
(15) Precipitation	PrecAvg	54.8	2.1	3.3	4.5	7.8	10.2	7.8	6.4	4.0	2.7	2.4	2.0	1.3	(15)	
	PrecMax	92.9	11.9	11.0	14.7	18.7	25.6	17.8	17.1	7.8	7.6	8.7	12.5	5.2	(16)	
	PrecMin	28.7	0.0	0.2	0.2	0.7	1.7	2.6	1.7	1.3	0.1	0.0	0.0	0.0	(17)	
	PrecStd	15.8	2.2	2.9	3.4	4.7	5.9	3.1	2.7	1.5	1.7	2.4	2.5	1.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.8	91.3	91.4	90.0	88.2	85.8	84.5	84.3	85.7	87.1	87.6	88.2	(19)	
		MCWB	80.4	80.5	80.8	79.5	79.3	77.5	76.2	75.9	77.7	78.2	79.7	79.4	(20)	
	2%	DB	88.2	89.7	89.7	88.2	86.4	84.4	82.8	82.7	84.3	86.0	86.2	87.7	(21)	
		MCWB	79.5	79.9	79.8	78.9	77.9	76.1	74.5	74.2	76.1	77.4	77.6	79.2	(22)	
	5%	DB	87.6	88.1	88.2	87.7	86.0	83.9	82.5	82.3	82.8	84.5	85.6	86.3	(23)	
		MCWB	79.1	79.2	79.1	78.7	77.5	75.7	74.2	73.8	74.6	76.1	77.1	77.7	(24)	
	10%	DB	86.2	87.4	87.5	86.2	84.5	82.6	81.0	80.9	82.4	84.1	84.4	85.8	(25)	
		MCWB	77.3	78.6	78.7	77.7	76.4	74.7	73.3	73.1	74.2	75.7	76.1	77.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.4	81.5	81.8	81.2	80.2	79.1	78.1	77.8	78.4	79.5	80.1	80.5	(27)	
		MCDB	87.7	89.2	89.0	86.3	85.2	83.1	82.0	81.6	83.3	85.4	85.8	86.9	(28)	
	2%	WB	80.1	80.5	80.8	80.2	79.5	78.0	77.1	76.6	77.4	78.2	78.9	79.7	(29)	
		MCDB	86.4	87.7	87.6	85.8	84.7	82.2	81.2	80.6	82.2	83.8	84.2	85.9	(30)	
	5%	WB	79.7	79.9	80.1	79.8	78.8	77.2	76.2	75.8	76.5	77.4	78.1	79.0	(31)	
		MCDB	85.6	86.6	86.4	85.3	84.0	81.5	80.3	79.7	81.1	82.7	83.2	84.9	(32)	
	10%	WB	79.0	79.3	79.5	79.2	78.1	76.4	75.3	75.0	75.8	76.7	77.3	78.2	(33)	
		MCDB	84.5	85.6	85.6	84.6	83.1	80.7	79.5	79.0	80.4	81.7	82.4	83.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	8.9	9.4	9.0	9.3	8.8	8.6	8.7	8.1	8.1	8.3	8.6	(35)	
		MCDBR	9.9	9.8	10.4	10.0	10.3	9.6	9.4	9.5	8.7	8.6	8.8	9.0	(36)	
	5% WB	MCWBR	4.6	4.4	4.9	4.5	5.0	5.0	4.9	4.9	4.4	4.2	4.3	4.3	(37)	
		MCWBR	9.8	9.7	10.0	9.3	9.7	9.0	9.0	8.7	8.6	8.7	8.7	8.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.390	0.386	0.383	0.379	0.378	0.369	0.370	0.377	0.411	0.412	0.403	0.403	(40)	
		taud	2.508	2.534	2.548	2.572	2.575	2.582	2.568	2.514	2.397	2.414	2.450	2.451	(41)	
	Ebn at Noon	Ebn at Noon	302	301	297	288	279	277	279	285	284	290	296	297	(42)	
		Edh at Noon	36	35	34	31	30	29	30	33	39	39	38	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2087	2037	1987	1768	1540	1458	1502	1677	1889	2039	2130	2149	(44)	
(45)		RadStd	150	95	92	104	109	81	60	78	85	80	129	88	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page